

Education

Massachusetts Institute of Technology

Expected May. 2030

PH.D. IN ELECTRICAL ENGINEERING AND COMPUTER SCIENCE (GPA: 5.0/5.0)

- Relevant Coursework: Deep Learning (A+), AI for Protein Biology (A), Advances in Computer Vision (A)

Tsinghua University

Graduated Jun. 2025

B.ENG. IN COMPUTER SCIENCE AND ENGINEERING (GPA: 3.94/4.0)

Industry Research Experience

Z.ai Student Researcher, Mentors: Zhenyu Hou, Yuxiao Dong & Jie Tang

Feb. 2025 - Jun. 2025

• Reinforcement Learning for Self-Improvement in Machine Learning Agents

- Designed an agent scaffold for long-horizon ML engineering with persistent memory to retain and build cumulatively on insights across past iterations, raising QwQ-32B's above-median rate on MLE-Bench-Lite from 30% to 47%.
- Formulated an RL training approach over single-step actions to leverage high-information trajectories, further increasing QwQ-32B's above-median rate on MLE-Bench-Lite from 47% to 56%.

BioMap Research Intern, Mentors: Bo Chen, Xingyi Cheng & Le Song

Aug. 2023 - Jun. 2024

• Scaling and Alignment of Protein Language Models (xTrimoPGLM, *Nature Methods* 2025)

- Fine-tuned multiple pre-training checkpoints of a 100B-parameter protein language model on 18 downstream tasks, uncovering four scaling behaviors, including grokking and inverse scaling.
- Improved conditional generation performance by 3 fold via supervised fine-tuning and reinforcement self-training.

• Protein Multiple Sequence Alignment (MSA) Generation (MSAGPT, *NeurIPS* 2024)

- Developed an autoregressive language model to generate synthetic MSAs for proteins in low-homology regimes, improving AlphaFold2 structure-prediction TM-score from 0.44 to 0.51 on orphan proteins in CASP.
- Designed and implemented AlphaFold2-guided RL for the MSA generator, using structure-prediction accuracy as the reward and further improving TM-score from 0.51 to 0.57.

Academic Research Experience

Massachusetts Institute of Technology

Ph.D. Researcher, Advisor: Prof. Yunha Hwang

Sep. 2025 - Present

• Inferring Bacterial Regulatory Networks from Genome Sequence (Hwang Lab)

- Predicting transcriptional regulatory interactions from evolutionary information in metagenomic sequences.
- Developing an interpretable deep learning model to understand and predict genomic perturbation effects.

• Protein Subcellular Localization Prediction (rotation with Prof. Kaiming He & Prof. Le Cong)

- Constructed a benchmark for protein subcellular localization classification and inpainting across 37 cell lines.
- Improved AUPRC by 22% over a sequence-only baseline on unseen cell lines by incorporating immunofluorescence microscopy images and transcriptomic profiles as cellular context.

• Protein Metal Binding Site Prediction (rotation with Prof. Yunha Hwang)

- Developed a sequence-based predictor of metal-binding residues for efficient metalloprotein screening.
- Enabled zero-shot generalization to unseen metal ions by learning a joint protein-metal embedding space.

• Few-shot Learning for Conditional Protein Generation (rotation with Prof. Alex Rives)

- Developed a masked diffusion language model that conditions on few-shot examples at inference time, achieving up to a 90% success rate in generating sequences with target EC classification across diverse enzyme families.
- Implemented classifier-free guidance during sampling, doubling the Spearman correlation between model-predicted residue probabilities and experimentally measured fitness relative to unguided sampling.

Carnegie Mellon University

Research Assistant, Advisor: Prof. Jian Ma

Jun. 2024 - Sep. 2025

• Reconstructing Missing Tissue Regions in Spatial Transcriptomics (Mimyr, *ICLR Gen2 Workshop* 2026)

- Developed a VQ-VAE tokenizer for gene-expression profiles and a Vision Transformer to reconstruct spatial transcriptomics data, achieving 23% higher Spearman correlation than a neighborhood-lookup baseline on held-out regions.

Tsinghua Knowledge Engineering Group

Advisors: Prof. Yuxiao Dong & Prof. Jie Tang

Nov. 2022 - May. 2023

• Diffusion-Based 3D Ligand Generation for Protein Binding Pockets

- Developed a protein-ligand affinity predictor to provide classifier guidance for a 3D ligand diffusion model.

- **Knowledge Graph-Based Therapeutic Target and Drug Discovery**

- Developed a graph-based framework combining knowledge-graph embeddings with community detection to prioritize candidate therapeutic targets and drugs for breast cancer.

Selected Publications

- Bo Chen*, **Zhilei Bei***, Xingyi Cheng, Pan Li, Jie Tang, Le Song. *MSAGPT: Neural Prompting for Protein Structure Prediction via MSA Generative Pre-Training*. *NeurIPS* (2024). * Equal contribution.
- Bo Chen*, Xingyi Cheng*, Pan Li, Yangli-ao Geng, Jing Gong, Shen Li, **Zhilei Bei**, et al. *xTrimoPGLM: Unified 100B-Scale Pre-Trained Transformer for Deciphering the Language of Proteins*. *Nature Methods* (2025). * Equal contribution.
- Ajinkya Deshpande, **Zhilei Bei**, Jian Ma, Spencer Krieger. *MIMYR: Generative Modeling of Missing Tissue in Spatial Transcriptomics*. *ICLR Generative AI in Genomics Workshop* (2026).

Technical Skills

ML Research	Representation learning, generative modeling, diffusion models, reinforcement learning
ML Systems	Distributed training, PyTorch, Hugging Face Transformers, DeepSpeed, SGLang
AI for Biology	Protein & genomic language models, spatial transcriptomics, small molecule & protein design
Programming	Python, C/C++, Rust, JavaScript, HTML/CSS, SQL, Assembly, VHDL

Honors & Awards

2025 **Sunlin (1966) and Priscilla Chou Fellowship**, MIT EECS Graduate Student Fellowship
2024 **Spark Scientific and Technological Innovation Fellowship**, Top 1% of Tsinghua University
2024 **Zhong Shimo Scholarship**, Top 1% of Computer Science Department, Tsinghua University
2024 **National Scholarship**, Top 3% of Tsinghua University
2020 **Finalist**, Regeneron International Science and Engineering Fair (ISEF)